I Wayan Agus Darmawan

+62 821-4670-9679 | darmawandeveloper@gmail.com | linkedin.com/in/agusdarmawnn | github.com/agus-darmawan | agus-darmawan.com

MASTER CV — internal reference only, no page limit. Not for direct submission; trim to a role-specific version (`software-developer/` or `robotics-engineer/`) before sending.

Professional Summary

Computer Engineering graduate (Institut Teknologi Sepuluh Nopember) with cross-disciplinary experience across full-stack web development and ROS-based robotics/embedded systems. Shipped 20+ freelance web projects using React, Next.js, Express, and AdonisJS, and led autonomous quadruped, ASV, and drone software programs from prototype through industrial field deployment. Comfortable owning a system end-to-end: hardware bring-up, control loops, backend services, and the operator-facing dashboards that tie it together.

Experience

PT. Ezra Robotics Teknologi May 2026 – Present
Robotics Systems Lead *Banten, Indonesia*

- Lead system integration and technical development for autonomous quadruped robots across industrial inspection, surveillance, and public-environment applications.
- Direct technical discussions, engineering coordination, and robotics system architecture spanning ROS-based software, localization, telemetry, and autonomous navigation.
- Facilitated a Lite3 Venture robot donation partnership with Universitas Sebelas Maret.

PT. Ezra Robotics Teknologi Nov. 2025 – May 2026
Robotics Engineer *Tangerang Selatan, Indonesia*

- Designed, developed, and tested quadruped inspection robots for electrical substations, including ROS-based applications, custom GUIs, and automated inspection features.
- Led outdoor localization and mapping improvements, resolving LIO challenges with RTK GPS for precise navigation in complex environments.
- Delivered field testing and technical demonstrations across PLN, Pertamina (oil & gas), and PTPN plantation deployments.

PT. Ezra Robotics Teknologi Dec. 2024 – Nov. 2025
Junior Robotics Engineer *Surabaya, Indonesia*

- Implemented base system integration on Deep Robotics Lite3 and X30 quadrupeds to support PPE-detection and indoor autonomous navigation.
- Developed ROS-based base programs integrating LiDAR-Inertial Odometry (LIO) and thermal imaging for substation inspection, plus a basic web-based telemetry/inspection monitor.

ASEEC Education Apr. 2023 – Mar. 2026
Research Mentor *Indonesia*

- Mentored middle and high school students for KOPSI, FIKSI, and national-level innovation competitions, guiding technical development and scientific writing from ideation to submission.
- Supported multidisciplinary student projects across IoT, machine learning, embedded systems, and web development, with several mentees returning across multiple cycles.

PT. ITS Tekno Sains Sep. 2024 – Aug. 2025
Robotics Software Engineer — iCar ITS *Surabaya, Indonesia*

- Developed autonomous driving software using ROS2, Fast-LIO2, and RTK GPS for reliable navigation in both GPS-available and GPS-denied environments.
- Engineered a web-based Ground Control Station using Next.js, ROSBridge, PostgreSQL, and Docker for real-time telemetry visualization, remote operation, and maintenance logging.
- Tuned PID controllers and optimized obstacle-avoidance algorithms to ensure vehicle stability during extensive outdoor testing.

B401 Robotics Laboratory Jun. 2024 – Aug. 2025
Laboratory Assistant *Surabaya, Indonesia*

- Engineered full-stack ROS-based autonomy and manipulation systems on physical TurtleBot and UR5 platforms, integrating perception, mapping, localization, motion planning, and control.
- Validated autonomous capabilities through real-world deployment and high-fidelity simulation (Gazebo, Isaac Sim, Webots).
- Led Digital Circuit laboratory modernization and managed a team of teaching assistants to improve practical learning outcomes.

Barunastra ITS Roboboat Team Dec. 2023 – Nov. 2024
Electrical & Programming Division Leader *Surabaya, Indonesia*

- Led the Electrical & Programming Division as a hands-on engineer, directing electrical system design, autonomous software development, and overall robot architecture with mechanical and non-technical teams.
- Designed and built end-to-end robotic platforms — electrical systems, sensor selection, control software, and integration — balancing mission requirements, performance, and budget for competition-ready deployments.

Barunastra ITS Roboboat Team Apr. 2022 – Dec. 2023
Robotics Software Engineer *Surabaya, Indonesia*
- Developed ROS-based autonomous navigation for an Autonomous Surface Vehicle (ASV), fusing LiDAR, camera, GPS, IMU, and compass data for perception, localization, and PID-based thruster control.
- Collaborated with electrical (low-level firmware) and computer-vision teams to field fully autonomous systems at KKI 2022, IRC 2023, and KKI 2023.

PT Satkomindo Mediyasa Feb. 2024 – Jun. 2024
Full Stack Developer *Jakarta Selatan, Indonesia*
- Built an internal Career Management Platform with Easy Apply and automated CV standardization, including candidate, recruiter, and admin dashboards with authentication, role-based access control (RBAC), API integration, and validation.
- Containerized the platform with Docker and shipped it through a sprint-based development workflow.
- Engineered backend services for a Document Tracking system in AdonisJS: authentication, role management, email verification, and workflow monitoring.
- Designed the concept architecture for an IoT-based NVR monitoring system tracking surveillance device uptime.

MAGE (Multimedia And Game Event) 9 Jan. 2023 – Feb. 2024
Frontend Developer *Surabaya, Indonesia*
- Built the official MAGE 9 ITS landing website with Next.js and Tailwind CSS — a first professional web project, self-taught while shipping.
- Took ownership of development when teammates became unavailable; maintained and deployed the production site with a responsive UI and Firebase authentication.

PT. ITS Tekno Sains Sep. 2023 – Dec. 2023
Robotics Software Lead — Autonomous Surface Vehicle (Project-Based) *Surabaya, Indonesia*
- Led software development for a ROS-based Autonomous Surface Vehicle (ASV) for hydrographic bathymetry surveys, a collaborative project between PT ITS Tekno Sains, PT Robomarine Indonesia (mechanical & electrical), and STTAL (end user).
- Designed and implemented autonomous navigation using Pure Pursuit with GPS RTK and IMU localization, integrated YOLO-based buoy detection, and developed an RViz-based mission control and monitoring interface.
- Directed software integration, mentored MBKM ITS engineers, led field trials, and resolved post-deployment issues to deliver a field-ready autonomous marine system.

PROJECTS

NextGen GCS | *Tauri, React, TypeScript, Go, PyTorch, TensorRT, NATS, PostgreSQL, CesiumJS* 2026
- Built a full-stack AI-powered ground control station for drone operations: a Tauri desktop app streaming live video (RTSP to WebRTC/HLS via MediaMTX) with real-time YOLO object detection and CesiumJS-based 3D flight tracking.
- Designed a Go backend coordinating a NATS message bus, PostgreSQL storage, and a Python/PyTorch/TensorRT inference service, paired with a custom OpenIPC/WFB-ng digital video link replacing the previous analog FPV setup.

Molly Dashboard | *Next.js 16, React, PostgreSQL, Auth.js, Zustand, roslib.js* 2026
- Built a robot-fleet operator console for the Deep Robotics Lite3 (Jetson Orin NX, ROS2 Humble): a live occupancy-map canvas with zoom/pan and POI tooltips, bridged to ROS2 via roslib.js with automatic reconnect.

GENISOM Quadruped SDK | *ROS2, Python, Jetson, Rockchip* 2026
- Reverse-engineered and extended the control SDK for a ZSBot GENISOM L1W EDU quadruped's dual-board (Firefly Rockchip + Jetson) architecture, diagnosing an undocumented fault mode and scoping a path to autonomous control via ROS2 Nav2.

Molly Controller | *ROS2, C++* 2026
- Wrote a production-grade ROS2 C++ package for the Jueying Lite3 with a four-layer transport-to-bridge architecture and a self-registering factory pattern for extending to additional robot backends.

Offline 3D Mapping Pipeline | *Docker, FAST-LIO2, LIO-SAM, Livox* 2026
- Built a Dockerized offline mapping workspace for the Unitree Go2 EDU (Livox Mid-360 + RealSense D435), benchmarking FAST-LIO2, LIO-SAM, and FAST-LIVO2.

ESP32 PID Motor Controller | *ESP-IDF, PlatformIO, C++* 2026
- Developed modular ESP32-C3 firmware converting PX4 AUX RC PWM into brushed-ESC speed control, with an anti-windup PID loop, encoder feedback, and a WiFi SoftAP web UI for live tuning.

Face Recognition Web | *FastAPI, React, TypeScript, MediaPipe, InsightFace* 2025

- Ported a Flutter/TFLite mobile face-recognition app to a FastAPI + React/TypeScript web stack using MediaPipe for detection and InsightFace (ArcFace) embeddings.

ROS2 Teaching Workspace | *ROS2 Humble, C++, Python* 2025

- Authored a 12-package, BSD-3-Clause ROS2 tutorial workspace covering topics, services, actions, parameters, TF2, and launch files, used in university lectures.

Bot BBM TOSIGA | *Google Apps Script, Cloudflare Workers* 2025

- Built and iterated (v1–v9) a Telegram bot that logs vehicle fuel data to Google Sheets, migrating from Apps Script to Cloudflare Workers to handle concurrent requests.

ROS2 Remote Dev Container | *Docker, NoMachine* 2025

- Built a parameterized Docker + NoMachine (NX) remote-desktop container for ROS2 development, resolving environment-variable and port-conflict bugs across machines.

Educational Robotics Kits (SD–SMA) 2026

- Developed the bill of materials and go-to-market pricing for an educational robotics-kit product line, including custom PCB/laser-cut fabrication costs and a reseller partnership.

PUBLICATIONS

Development of a Quadruped-Legged Robot for Position Estimation of Overheat Components in Electrical Substations Using a Thermal Camera

Hand Gesture Recognition for Collaborative Robots Using Lightweight Deep Learning in Real-Time Robotic Systems

HONORS & AWARDS

Grand Champion — International Roboboat Competition (IRC) 2023

1st Place, National Unmanned Water Vessel Competition (KKI) — 2022 & 2023

2nd Place, Design Documentation — International Roboboat Competition (IRC) 2023

1st Winner, Paper Competition & Favorite Paper Award — ICAVETS 2022 (International)

TECHNICAL SKILLS

Languages: C/C++, Python, TypeScript/JavaScript, Swift, SQL
Web & Full-Stack: React, Next.js, Tailwind CSS, Zustand, Express, AdonisJS, FastAPI, Node.js, REST APIs, Auth.js, RBAC
Robotics & Embedded: ROS/ROS2, rosbridge/roslib.js, PID control, sensor fusion, SLAM (FAST-LIO2, LIO-SAM), Livox/LiDAR, RTK GNSS, MAVLink, PlatformIO/ESP-IDF
Simulation & Platforms: Gazebo, Isaac Sim, Webots, quadrupeds, ASVs, AGVs, quadcopters
AI & Vision: YOLO, PyTorch, TensorRT, MediaPipe, InsightFace
DevOps & Data: Docker, Git, GitHub Actions CI/CD, Linux, PostgreSQL, SQLite, Cloudflare Workers, NATS

EDUCATION

Institut Teknologi Sepuluh Nopember (ITS) Surabaya, Indonesia
Bachelor of Engineering in Computer Engineering *Jul. 2021 – Sep. 2025*

Apple Developer Academy | **Indonesia** Surabaya, Indonesia
iOS Developer Cohort *Feb. 2025 – Dec. 2025*